

EXP 3 Deffie hellman key exchnage

Server.java

```
import java.io.*;
import java.net.*;
import java.math.BigInteger;
import java.util.Random;

public class Server {

    public static void main(String[] args) throws Exception {

        ServerSocket ss = new ServerSocket(5000);
        System.out.println("Server Waiting...");

        Socket s = ss.accept();

        BufferedReader in =
            new BufferedReader(
                new InputStreamReader(s.getInputStream()));

        PrintWriter out =
            new PrintWriter(s.getOutputStream(), true);

        BigInteger p = new BigInteger(in.readLine());
        BigInteger g = new BigInteger(in.readLine());

        Random r = new Random();

        BigInteger b = new BigInteger(5, r);

        BigInteger B = g.modPow(b, p);

        out.println(B);

        BigInteger A = new BigInteger(in.readLine());

        BigInteger key = A.modPow(b, p);

        System.out.println("Server Private Key = " + b);
        System.out.println("Server Public Key = " + B);
        System.out.println("Shared Secret Key = " + key);

        s.close();
        ss.close();
    }
}
```

```

Client.java
import java.io.*;
import java.net.*;
import java.math.BigInteger;
import java.util.*;

public class Client {

    public static void main(String[] args) throws Exception {

        Socket s = new Socket("localhost", 5000);

        BufferedReader in =
            new BufferedReader(
                new InputStreamReader(s.getInputStream()));

        PrintWriter out =
            new PrintWriter(s.getOutputStream(), true);

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Prime Number (p): ");
        BigInteger p = new BigInteger(sc.nextLine());

        System.out.print("Enter Generator (g): ");
        BigInteger g = new BigInteger(sc.nextLine());

        out.println(p);
        out.println(g);

        Random r = new Random();

        BigInteger a = new BigInteger(5, r);

        BigInteger A = g.modPow(a, p);

        BigInteger B = new BigInteger(in.readLine());

        out.println(A);

        BigInteger key = B.modPow(a, p);

        System.out.println("\nClient Private Key = " + a);
        System.out.println("Client Public Key = " + A);
        System.out.println("Server Public Key = " + B);
        System.out.println("Shared Secret Key = " + key);

        s.close();
    }
}

```